



Examination Number:

Set:

Shore

Year 12

HSC Assessment Task 3

Half-Yearly Exam

April 27 2015

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided separately
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- In Questions 11–14 show relevant mathematical reasoning and/or calculations
- Start each of Questions 11–14 in a new writing booklet
- Write your examination number on the front cover of each booklet

If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Total marks – 70

Section I Pages 2–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–12

60 marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1 The polynomial $2x^3 + x - 4 = 0$ has roots α, β and γ . What is the value of $\alpha + \beta + \gamma$?

(A) -2

(B) $-\frac{1}{2}$

(C) 0

(D) 2

- 2 The point R divides the interval $P(3, -6)$ and $Q(6, -9)$ externally in the ratio 2:1.

What are the coordinates of R ?

(A) $(9, -12)$

(B) $(5, -8)$

(C) $(4, -7)$

(D) $(0, -3)$

- 3 If $\int_0^k (3x - 6) dx = 0$ and $k \neq 0$, what is the value of k ?

(A) 4

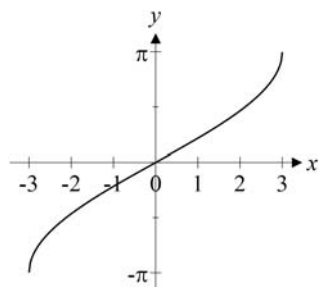
(B) 2

(C) -2

(D) -4

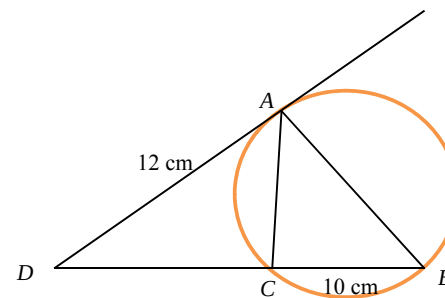
DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

- 4 What is the equation of the function shown in the graph below?



- (A) $y = \frac{1}{2} \sin^{-1} 3x$
- (B) $y = \frac{1}{2} \sin^{-1} \frac{x}{3}$
- (C) $y = 2 \sin^{-1} \frac{x}{3}$
- (D) $y = 2 \sin^{-1} 3x$
- 5 Which of the following is the Cartesian Equation of the variable point $P(2\cos t, 2\sin t)$, where t is the parameter?
- (A) $y = \tan x$
- (B) $x^2 + y^2 = 1$
- (C) $x^2 + y^2 = 4$
- (D) $x^2 = 4ay$

- 6 ABC is a triangle inscribed in a circle. The tangent to the circle at A meets BC produced at D where $BC = 10$ cm and $AD = 12$ cm. What is the length of BD ?



NOT TO SCALE

- (A) 8
- (B) 18
- (C) 26
- (D) 28
- 7 Let $t = \tan \frac{\theta}{2}$ where $0 < \theta < \pi$. Which of the following gives the correct expression for $\sin \theta - \cos \theta$?
- (A) $\frac{2t - 1 - t^2}{1 + t^2}$
- (B) $\frac{t^2 + 2t + 1}{1 + t^2}$
- (C) $\frac{1 - t^2 - 2t}{1 + t^2}$
- (D) $\frac{t^2 + 2t - 1}{1 + t^2}$

8 What is the value of $\cos 2\theta$, given $\cos \theta = \frac{3}{5}$ and $\sin \theta \geq 0$?

- (A) $\frac{6}{5}$
- (B) $\frac{24}{25}$
- (C) $\frac{7}{25}$
- (D) $-\frac{7}{25}$

9 Which of the following is an expression for $\frac{d}{dx}\left(\tan^{-1} \frac{x}{2}\right)$?

- (A) $\frac{2}{2+x^2}$
- (B) $\frac{2}{4+x^2}$
- (C) $\frac{4}{2+x^2}$
- (D) $\frac{4}{4+x^2}$

10 Which of the following is the solution to the expression $\int \frac{dx}{\sqrt{9-4x^2}}$?

- (A) $\sin^{-1} \frac{2x}{3} + c$
- (B) $\sin^{-1} \frac{3x}{2} + c$
- (C) $\frac{1}{2} \sin^{-1} \frac{2x}{3} + c$
- (D) $\frac{1}{2} \sin^{-1} \frac{3x}{2} + c$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Start each of Questions 11–14 in a new writing booklet.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{4x}$. 1

(b) Find $\frac{d}{dx} \left(\frac{\sin 3x}{x} \right)$. 2

(c) Find the size of the acute angle between the lines $y - 5x - 9 = 0$ and $3y = 2x + 8$. 2

(d) Find $\int (1 + \tan^2(x+1)) dx$. 2

(e) Find in simplest form the exact value of $\cos 15^\circ$. 2

(f) Solve $\frac{2}{x+3} \geq 1$. 3

(g) The area of a sector of a circle of radius 9 cm is 75 cm^2 . Find the length of the arc of the sector. 3

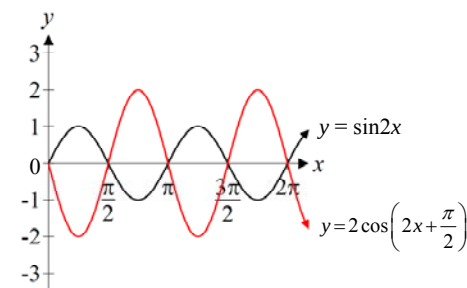
Question 12 (15 marks) Use a SEPARATE writing booklet

(a) Find the values of a and b that make the polynomial $P(x) = 2x^3 + ax^2 - 13x + b$ exactly divisible by $x^2 - x - 6$. 3

(b) (i) Express $\sin x - \sqrt{3} \cos x$ in the form $R \sin(x - \theta)$ where $R > 0$ and $0 < \theta < \frac{\pi}{2}$. 2

(ii) Hence or otherwise solve the equation $\sqrt{2} = \sin x - \sqrt{3} \cos x$ for $0 \leq x \leq 2\pi$. 2

(c) The diagram below shows two curves $y = \sin 2x$ and $y = 2 \cos\left(2x + \frac{\pi}{2}\right)$. 3

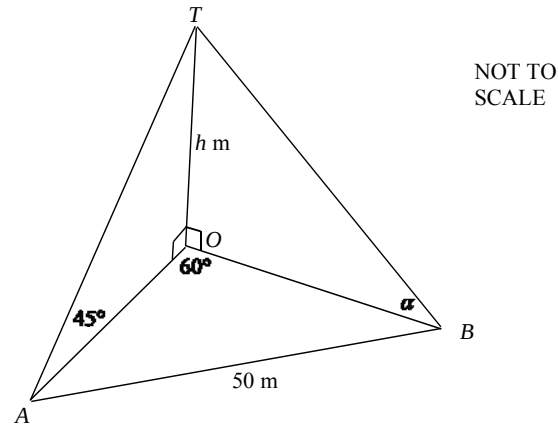


Determine the area enclosed between the curves for the domain $0 \leq x \leq \pi$.

Question 12 continued on page 9

Question 12 (continued)

(d)



In the diagram, the points A , B and O are in the same horizontal plane. A and B are 50 m apart and $\angle AOB = 60^\circ$. OT is a vertical tower of height h metres.

The angles of elevation of T from A and B respectively are 45° and α . (α is acute).

- | | |
|--|---|
| (i) Explain why $AO = h$. | 1 |
| (ii) Prove that $h^2 \cot^2 \alpha - h^2 \cot \alpha + h^2 = 50^2$ | 2 |
| (iii) Given the tower is 30 m high, find the angle α correct to the nearest degree. | 2 |

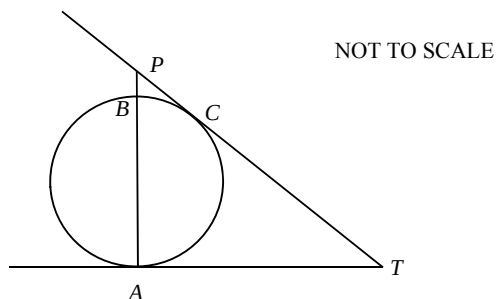
Question 13 (15 marks) Use a SEPARATE writing booklet

- | | |
|---|---|
| (a) (i) Find $\frac{d}{dx}((\log_e(x+4))^3)$. | 2 |
| (ii) Hence or otherwise evaluate $\int_{-3}^0 \frac{(\log_e(x+4))^2}{x+4} dx$. | 2 |
| (b) A straight line through $T(0, -a)$ cuts the parabola $x^2 = 4ay$ at $P(2ap, ap^2)$ and $Q(2aq, aq^2)$. | |
| (i) Show that the equation of TP is | 2 |
| $2py = x(p^2 + 1) - 2ap$. | |
| (ii) Prove that for TP to pass through Q , $pq = 1$. | 2 |
| (iii) Hence or otherwise prove that $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$, where S is the focus of the parabola. | 3 |

Question 13 continued on page 11

Question 13 (continued)

- (c) In the diagram, AB is a diameter of the circle ABC . The tangents at A and C meet at T . The lines TC and AB are produced to meet at P .



Copy the diagram into your examination booklet.

- (i) Prove that $\angle BCP = 90^\circ - \angle CAT$. 2
- (ii) Explain why $ATCB$ could never be a cyclic quadrilateral. 2

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) Use Mathematical Induction to prove that 3
 $1 \times 2 + 2 \times 2^2 + 3 \times 2^3 + \dots + n \times 2^n = (n-1) \times 2^{n+1} + 2$ for $n \geq 1$.
- (b) Consider the function $f(x) = \frac{e^x}{4+e^x}$.
- (i) Determine whether $f(x)$ has any stationary points. 3
- (ii) Find the point of inflexion given that 2
 $f''(x) = \frac{4e^x(4-e^x)}{(4+e^x)^3}$.
- (iii) Show that $0 < f(x) < 1$ for all x . 2
- (iv) Sketch the curve $y = f(x)$. 2
- (v) Explain whether $f(x)$ has an inverse function or not. 1
- (vi) Find the inverse function $y = f^{-1}(x)$. 2

END OF PAPER

- Q1 C Q6 B
Q2 A Q7 D
Q3 A Q8 D
Q4 C Q9 B
Q5 C Q10 C

11a) $\lim_{x \rightarrow 0} \frac{\sin 5x}{4x} = \frac{1}{4} \lim_{x \rightarrow 0} \frac{\sin 5x}{x}$
 $= \frac{5}{4} \lim_{x \rightarrow 0} \frac{\sin 5x}{5x}$
 $= \frac{5}{4}$

b) $\frac{d}{dx} \left(\frac{\sin 3x}{x} \right) = \frac{3x \cos 3x - \sin 3x}{x^2}$

c) $y = 5x + 9 \quad m = 5$
 $3y = 2x + 8 \quad m = \frac{2}{3}$

$\tan d = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{5 - \frac{2}{3}}{1 + \frac{10}{3}} \right|$
 $= 1$

$d = 45^\circ$

d) $\int (1 + \tan^2(x+1)) dx$
 $= \int \sec^2(x+1) dx$
 $= \underline{\underline{\tan(x+1) + C}}$

- Q6 B
Q7 D
Q8 D
Q9 B
Q10 C

e) $\cos 15^\circ = \cos(45^\circ - 30^\circ)$
 $= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2}$
 $= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$
 $= \frac{\sqrt{3} + 1}{2\sqrt{2}}$

f) $\frac{2}{x+3} \geq 1 \quad x \neq -3$
 $2(x+3) \geq (x+3)^2$
 $0 \geq (x+3)(x+1)$

$-3 < x \leq -1$

g) Area = $75 \text{ cm}^2 \quad r = 9$

$A = \frac{1}{2} r^2 \theta$
 $75 = \frac{1}{2} \times 81 \times \theta$
 $\theta = \frac{150}{81}$

$l = r\theta$
 $= 9 \times \frac{150}{81}$
 $= \underline{\underline{16\frac{2}{3} \text{ cm}}}$

12a) $P(x) = 2x^3 + ax^2 - 13x + b$
 $(x-3)$ and $(x+2)$ are factors
 $P(3) = 54 + 9a - 39 + b = 0$
 $P(-2) = -16 + 4a + 26 + b = 0$

$0 = 10 + 4a + b$

$0 = 15 + 9a + b$

$0 = 5 + 5a$

$\underline{\underline{a = -1}}$

$10 - 4 + b = 0$

$\underline{\underline{b = -6}}$

b) i) $\sin x - \sqrt{3} \cos x = R \sin(x - \alpha)$
 $\sin x - \sqrt{3} \cos x = R \sin x \cos \alpha - R \cos x \sin \alpha$

$R \cos \alpha = 1$

$R \sin \alpha = \sqrt{3}$

$R^2 = 1^2 + \sqrt{3}^2$

$R = 2 \quad R > 0$

$\tan \alpha = \sqrt{3}$

$\alpha = \frac{\pi}{3}$

$\sin x - \sqrt{3} \cos x = 2 \sin(x - \frac{\pi}{3})$

ii) $\sqrt{2} = 2 \sin(x - \frac{\pi}{3})$
 $\frac{1}{\sqrt{2}} = \sin(x - \frac{\pi}{3})$

$x - \frac{\pi}{3} = \frac{\pi}{4} \text{ and } \frac{3\pi}{4}$

$\underline{\underline{x = \frac{7\pi}{12} \text{ and } \frac{13\pi}{12}}}$

c) $A = 2 \int_0^{\frac{\pi}{2}} (\sin x - 2 \cos(2x + \frac{\pi}{2})) dx$
 $= 2 \left[-\frac{1}{2} \cos x - \sin(2x + \frac{\pi}{2}) \right]_0^{\frac{\pi}{2}}$
 $= 2 \left(\frac{1}{2} - (-1 - 1) \right)$
 $= \underline{\underline{6}}$

d) i) $AO = h$ as $\triangle ATO$ is isosceles
as $\angle ATO = 45^\circ \therefore AO = OT$

ii) $AO = h \quad \frac{h}{OB} = \tan d$
 $OB = h \cot d$

$a^2 = b^2 + c^2 - 2bc \cos A$
 $50^2 = h^2 + h^2 \cot^2 d - 2h \times h \cot d \cos 60^\circ$
 $= h^2 + h^2 \cot^2 d - 2h^2 \cot d \times \frac{1}{2}$
 $50^2 = h^2 + h^2 \cot^2 d - h^2 \cot d$

iii) $50^2 = 30^2 + 30^2 \cot^2 d - 30^2 \cot d$
 $1600 = 900 \cot^2 d - 900 \cot d$
 $0 = 9 \cot^2 d - 9 \cot d - 16$

Let $u = \cot d$
 $0 = 9u^2 - 9u - 16$

$u = \frac{9 \pm \sqrt{81 - 4 \times 9 \times -16}}{2 \times 9}$

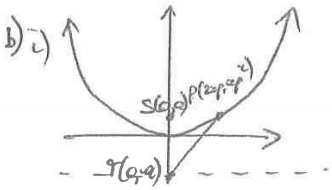
$= 1.924 \quad u > 0$

$\tan d = \frac{1}{1.924}$

$\underline{\underline{d = 27^\circ}}$

13x) i) $\frac{d}{dx} (\ln(x+4))^3 = \frac{3(\ln(x+4))^2}{x+4}$

$$\begin{aligned} \text{ii)} \quad & \int_{-3}^0 \frac{(\ln(x+4))^2}{x+4} dx \\ &= \frac{1}{3} \left[(\ln(x+4))^3 \right]_{-3}^0 \\ &= \frac{1}{3} \left[(\ln 4)^3 - 0 \right] \\ &= \frac{1}{3} \left[\underline{\underline{(\ln 4)^3}} \right] \end{aligned}$$



$$PT \Rightarrow m = \frac{a_p^2 + a}{2a_p} = \frac{p^2 r^1}{1 a_p}$$

$$b = -a$$

$$y = \pi x^2$$

$$y = \frac{p^2}{4\pi} x - a$$

$$2px = \frac{(p^2)}{4\pi} x - 2ap$$

ii) $\mathcal{Q}(2ap, ap^2)$ satisfies eqn

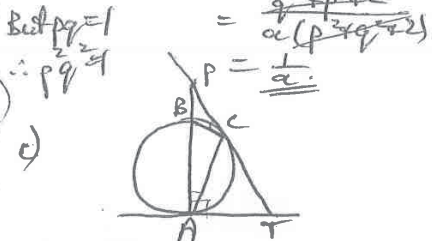
$$2apq^2 = (p^2u)2ap - 2ap$$
$$2apq^2 = 2ap^3 - 2ap$$
$$pq^2 - qp^2 = q - p$$
$$pq(q-p) = q-p$$
$$\underline{\underline{pq = 1}}$$

$$\text{iii) } \frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$$

SP $\Rightarrow$ by defn of Parabola
 $= ap^2 + a$

$$SQ = aq^2 + a$$

$$\begin{aligned} \therefore \frac{1}{SP} + \frac{1}{SQ} &= \frac{1}{a(p^2+1)} + \frac{1}{a(q^2+1)} \\ &= \frac{q^2+1+p^2+1}{a(p^2+1)(q^2+1)} \\ &= \frac{q^2+p^2+2}{a(p^2q^2+p^2+q^2+1)} \\ &= \frac{q^2+p^2+2}{a(p^2q^2+p^2+q^2+2)} \\ \therefore p^2q^2 &= 1 \\ \therefore p^2q^2 &= 1 \end{aligned}$$



c) $\angle CAT = \angle APT$ $\angle CAT = 90^\circ$ (L between ~~TC~~ AT tangent & radius)

$\angle BAC = \angle BCP$ ($\angle$ between tangent & chord = $\angle$ in alternate segment)

$$\therefore \angle BAC = 90^\circ - \angle CAT$$

$$\therefore \angle BCP = 90^\circ - \angle CAT$$

ii) For cyclic quadrilateral
 $\angle BAT + \angle BCT = 90^\circ$
 but $\angle BCA = 90^\circ$ (in semicircle)
 $\angle BAT = 90^\circ$
 $\therefore \angle BCT = \angle BCA + \angle ACT$
 $\angle BAT + \angle BCT = 90^\circ + 90^\circ + \angle ACT$
 $\neq 180^\circ$

$$14.9) \quad 1x2 + 2x2^2 + \dots + nx2^n = (n-1)x2^{n+1} + 2$$

(Si) Prove true for $n=1$
 $1 \times 2 = 0 \times 2^2 + 2 \checkmark$
 True for $n=1$

⑤ Assume true for $n=k$
 $1 \times 2 + 2 \times 2^2 + \dots + k \times 2^k = (k-1)2^{k+1} + 2$

(S3) Prove true for $n=k+1$
 $1 \times 2 + 2 \times 2^2 + \dots + k \times 2^k + (k+1) \times 2^{k+1} = k \times 2^{k+2} + 2^{k+2}$

From Step 2

$$1 \times 2 + 2 \times 2^2 + \dots + k \times 2^k = (k-1)2^{k+1} + 2$$

$$\begin{aligned} \therefore 1 \times 2 + 2 \times 2^2 + \dots + k \times 2^k + (k+1) \times 2^{k+1} \\ = (k+1) \times 2^{k+1} + (k+1) \times 2^{k+1} + 2 \\ = 2^{k+1} (k+1 + 1) + 2 \\ = 2^{k+1} \cdot 2k + 2 \\ = k \cdot 2^{k+2} + 2. \end{aligned}$$

(S4) Assuming true for $n=1$ & $n=k+1$,
assuming true for $n=k$,
let $k=1 \therefore n=2$
 $k=2 \therefore n=3$ etc
 $\therefore$ true for all integral n
 x

$$b) i) y = \frac{e^x}{4+e^x}$$

Let Pts at $\frac{dy}{dx} = 0$

$$y' = \frac{e^x(4+e^x) - e^x \cdot e^x}{(4+e^x)^2}$$

$$= \frac{4e^x + e^{2x} - e^{2x}}{(4+e^x)^2}$$

$$0 = 4e^x$$

$\therefore c^x \neq 0 \therefore$ Not possible.

∴ No stat A/s

ii) Possible pt of inflexion at $x=2$

$$y''' = 0$$

$$y'' = \frac{4e^x(4e^x)}{(4e^x)^3}$$

$$0 = 4 - e^x$$

$$x = \underline{\ln 4}, \quad \therefore y = \frac{4}{4+4} = \underline{\underline{\frac{1}{2}}}$$

$$\therefore (2n^4, \frac{1}{2})$$

Test for pt of inflexion

x	$\ln 3$	$\ln 4$	$\ln 5$
y''	$\frac{12}{7^3}$	0	$-\frac{20}{9^3}$

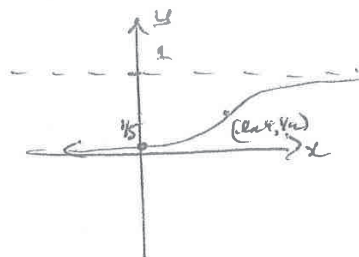
as sign of y'' changes

$\therefore (ln 4, \frac{1}{2})$ is a pt of inflexion

$$(iii) \quad \lim_{x \rightarrow \infty} \frac{e^x}{4+e^x} = \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{4}{e^x}} = \frac{1}{1+0} = 1$$

$$\lim_{x \rightarrow \infty} \frac{e^x}{4e^x} = 0.$$

iv)



i) Always ~~is~~ monotonic increasing or
horizontal line test

$$ii) y = \frac{e^x}{e^x + 4}$$

$$x = \frac{e^y}{e^y + 4}$$

$$xe^y + 4x = e^y$$

$$(x-1)e^y = -4x$$

$$e^y = \frac{4x}{1-x}$$

$$y = \underline{\underline{\ln\left(\frac{4x}{1-x}\right)}}$$